Reg. No.:

Name :

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act 1956)

Continuous Assessment Test II– March 2025

Programme	:	B.Tech (CSE and Specialization)	Semester	:	Winter 2024-2025
Course Code	:	BCSE307L	Class Number	:	CH2024250502764, CH2024250501805, CH2024250501806, CH2024250501807, CH2024250502336, CH2024250502338, CH2024250502763
Course Title	:	Compiler Design	Slot	:	B1+TB1
Faculty	:	Prof. Anand M, Prof.D.Jeya Mala, Prof. Shivani Gupta, Prof, Manju G, Prof. Sureshkumar, Prof.Senthil Prakash, Prof.Mercy Rajaselvi Beaulah			
Time	:	90 Minutes	Max. Marks	:	50

Answer ALL Questions

Q. No.	Question Text	Marks
1.	"XYZ Hospitals" developed a "Pacemaker Machine" for cardiovascular patients. For the functioning of the firmware operations, a set of grammar rules are created and the machine has to take the readings from the auricle and ventricle and find the difference between them ((ie.) hv = av – vv *2), where hv is final heart reading, av is auricle reading, vv is ventricle reading) to do certain internal decision making. The grammar rules are given below: 1. **S → A** 2. **A → V = E** 3. **E → E+T | E-T | T** 4. **T → T*F** 5. **F → id | (E) | digit** Do the following: a) Generate the LR(0) canonical collection for the given grammar (5 marks). b) Construct the SLR(1) parsing table (5 marks). c) Assume that, you are involved in this process, prove that the operation string hv = av – vv *2 is accepted using SLR parsing.(5 Marks).	15
2.	(a) For a communication network, it is required to check whether the receiver has received the same data or any loss of data occurred or any virus data is passed along with it as per the grammar rules given below. 1. **M → SR** 2. **S → BS / B** 3. **R → BR / B** 4. **B → 1/0** The data is represented in binary format. In order to do the check, you are asked to (i) provide SDD with SDT using attribute grammar with S-attributes and L-attributes such as length, count, val, etc. (5 Marks)	15

	(ii) Perform attribute evaluation using annotated parse tree to check the length of a sample message – 1110. (5 Marks) ---------- (b) Consider a text string, that is, a combination of only the characters 'a','b' and 'd'. Prepare production rules, semantic rules, SDT with L-attributes and final syntax tree for such string creation. Also, check whether the string "abbcd".is accepted or not using the tree. (5 Marks)	

3.	(a) Consider the below code snippet in C, showcase the three address code generated by the compiler during intermediate code generation phase: (10 Marks)
	int i,j,n,flag;
	int sum = 0;
	for (int i = 0; i < 10; i++)
	{ scanf("%d", &n);
	if (n >=0 && n <=2) sum +=n;
	else
	{
	for(j=2; j< n/2 ; j++)
	{
	if (n % j !=0) { flag=1; continue;}
	else {flag=0; break;}
	}
	if (flag ==1) sum += n;
	}
	printf(%d".sum);
	}
	(b) Prepare a parse tree for the same. (5 Marks)
		15

4.	For the arithmetic expression x =(p-q) * (r/s) / t^r (a) Construct Three Address Code (3AC) and represent them using Quadruples and Indirect Tuples: (3 Marks) (b) Construct DAG representation of syntax tree. (2 Marks) __****__	5